

MedGraphRAG: A Safety-First Hybrid Graph Retrieval-Augmented Generation System for Offline Clinical Decision Support

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Abstract

Background and Objective. Large language models (LLMs) deployed in clinical decision support face a safety-utility trade-off: unconstrained answering yields unacceptable rates of wrong assertions, whereas excessive refusal renders systems inert. Existing retrieval-augmented generation (RAG) frameworks lack mechanistic, bounded safety controls deployable on institutional hardware. We present MedGraphRAG, an offline hybrid GraphRAG system that reframes clinical AI evaluation from unconstrained accuracy maximisation to bounded wrong-assertion rate (WAR).

Methods. MedGraphRAG couples dense vector retrieval (FAISS IVF-PQ with MedCPT embeddings) and lexical retrieval (BM25) with an embedded property graph (Kuzu) enriched with 632,931 relational edges (617,219 NEGATES and 15,712 TEMPORAL_BEFORE). Dual candidate streams are merged via Reciprocal Rank Fusion and re-ranked with an additive relation-aware score ($\gamma = 0.15$). A beta-gated verification layer enforces a strict $\text{WAR} \leq 10\%$ constraint by combining evidence faithfulness (φ) and graph consistency (S_g): $V = \beta\varphi + (1 - \beta)S_g$. The system executes entirely offline on a commodity laptop (Ryzen 7 7840HS / RTX 3050 6 GB) using Qwen2.5-7B-Instruct Q4_K_M. Private health data are encrypted with AES-256-GCM and HKDF-derived keys. Retrospective evaluation was performed on MedQA-US ($N = 500$, seed 42, $T = 0.0$).

Results. At the calibrated safe threshold ($\tau^* = 0.80$), Graph-Only mode (M2) achieves 60.8% precision-on-answered with $\text{WAR} = 8.0\%$ (79.6% refusal). Disabling the gate drives WAR to 44.6% across all 500 queries.

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Deterministic query rewriting expands answered coverage by 3.2–3.4 pp while maintaining bounded error. Hybrid retrieval completes in 455.7 ms (95% CI: [388.4, 495.8] ms); full end-to-end pipeline latency is 4,083 ms (M2) to 12,667 ms (M4).

Conclusion. MedGraphRAG demonstrates that bounded clinical error and interpretable offline decision support are jointly achievable on consumer hardware without cloud dependencies or GPU clusters.

Keywords: GraphRAG, clinical NLP, hallucination mitigation, knowledge graph, retrieval augmented generation, decision support, safety

1. Introduction

Deploying large language models (LLMs) as point-of-care clinical assistants offers direct utility for diagnostic and therapeutic workflows [1, 2]. However, the tendency of ungrounded parametric architectures to hallucinate plausible yet factually incorrect assertions poses direct patient-safety hazards [3]. In high-stakes medicine, confident assertions regarding contraindicated medications, misidentified diagnostic criteria, or inverted pathological chronologies can cause serious clinical harm.

To mitigate hallucinations, retrieval-augmented generation (RAG) grounds language generation on passages retrieved from external corpora [4, 5]. Standard medical RAG frameworks rely on dense bi-encoders [6, 7], which match semantic similarity effectively but struggle with complex clinical syntax. Specifically, vector embeddings frequently compress or lose negation scope (e.g., conflating “no evidence of pulmonary embolism” with “pulmonary embolism”) and fail to enforce multi-step temporal chronologies required in differential diagnosis [8, 9].

Knowledge graphs provide a complementary symbolic representation, structuring clinical entities and ontological dependencies into explicit triples

19 [10, 11]. Recent hybrid GraphRAG systems combine graph structures with
20 textual retrieval [12–14] (Wu et al. share the MedGraphRAG name but focus
21 on entity-centric graph construction; our architecture emphasises hybrid
22 retrieval fusion and dual-gated verification). Nevertheless, existing medical
23 GraphRAG implementations exhibit three fundamental limitations:

24 First, current evaluations focus almost exclusively on unconstrained ac-
25 curacy (e.g., aggregate multiple-choice scores), omitting explicit bounds on
26 incorrect statements. In clinical practice, an assistant that answers 60% of
27 questions correctly while making wrong assertions on 40% is clinically un-
28 acceptable. Safe deployment requires a constrained-optimisation framework
29 that caps the wrong-assertion rate (WAR) below an acceptable tolerance
30 ϵ , treating refusal (abstention) as an active safety mechanism rather than
31 system failure.

32 Second, existing biomedical knowledge graphs capture predominantly pos-
33 itive ontological relations (e.g., TREATS or CAUSES), neglecting negation
34 and temporal progression. Without these primitives, retrieval graphs cannot
35 detect when retrieved passages contain superseded diagnoses or explicit ex-
36 clusions.

37 Third, most GraphRAG systems rely on cloud-hosted language models
38 (e.g., GPT-4) and server clusters. This architecture violates patient data pri-
39 vacy regulations (e.g., HIPAA, GDPR) that prohibit transmitting protected
40 health information (PHI) beyond institutional boundaries. Practical point-
41 of-care deployment requires secure, sub-second hybrid retrieval and inference
42 executing locally on commodity hardware.

43 We present **MedGraphRAG**, an offline hybrid GraphRAG system de-

44 signed for point-of-care clinical decision support. Our primary contributions
45 are:

- 46 1. **Safety-Constrained Pareto Calibration:** A verification framework
47 optimising answered accuracy subject to $\text{WAR} \leq 10\%$. Across 15
48 threshold settings on MedQA-US ($N = 500$), we establish the Pareto
49 frontier, demonstrating that ungated generation yields unacceptable
50 error rates ($\approx 44\%$).
- 51 2. **Relational Graph Enrichment:** Ingestion of 632,931 negation and
52 temporal relational edges into an embedded Kuzu graph, demonstrat-
53 ing mechanistically how symbolic consistency checks overturn halluci-
54 nations bypassing dense retrieval.
- 55 3. **Deterministic Query Rewriting and**
56 **Relation Reranking:** A temperature-zero query normaliser improv-
57 ing answered coverage by 3.2–3.4 pp at matched thresholds, combined
58 with a relation-aware reranker boosting Graph-Hit@5 by 13.6 pp.
- 59 4. **Hardened Offline Stack on Consumer Hardware:** A self-
60 contained system operating on an NVIDIA RTX 3050 (6 GB) laptop
61 GPU, achieving hybrid retrieval in 455.7 ms with authenticated
62 AES-256-GCM encryption and a resolved 13-point security audit.

63 Supplementary materials provide the DECIDE-AI applicability mapping,
64 IJMI ML checklist, and extended tables; configuration details appear in Ap-
65 pendix A.

66 **2. Methods**

67 *2.1. System Architecture*

68 Figure 1 shows the MedGraphRAG inference pipeline. Stage 1 determin-
69 istically standardises input queries into clinical phrasing. Stage 2 executes
70 dual-stream dense and lexical retrieval, combining candidates via Reciprocal
71 Rank Fusion and relation-aware graph reranking. Stage 3 expands clinical
72 entities within the embedded Kuzu graph. Stage 4 computes composite beta-
73 gated verification scores, triggering structured refusals for low-confidence
74 queries. Stage 5 generates evidence-grounded responses using a quantised
75 LLM isolated behind a GPU mutex lock.

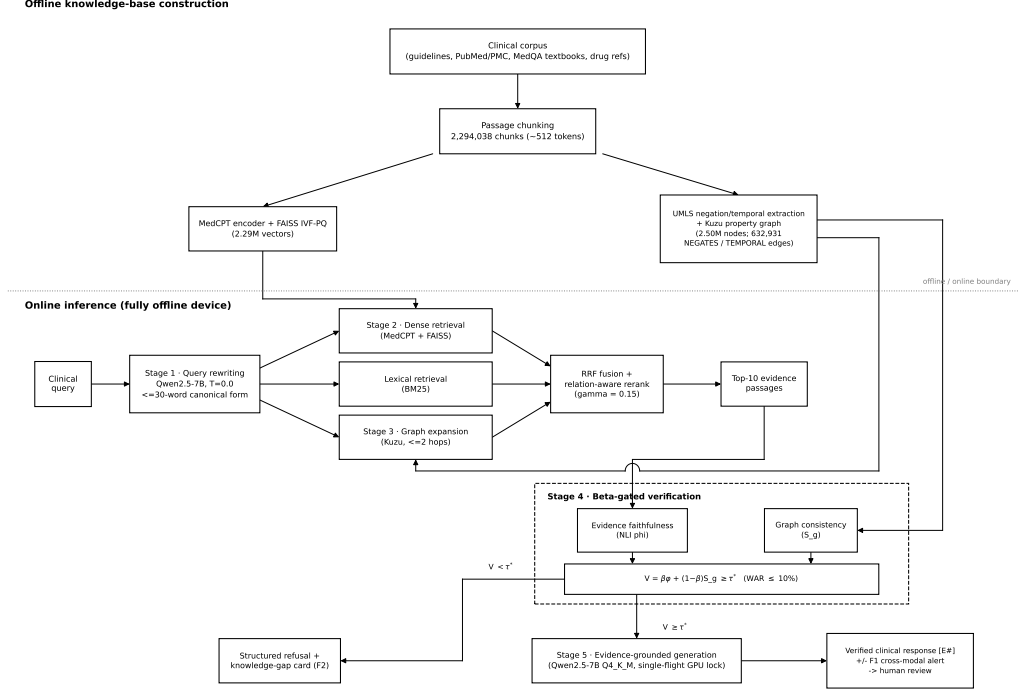


Figure 1: MedGraphRAG five-stage offline pipeline architecture. Stage 1 executes deterministic query rewriting ($T = 0.0, \leq 30$ words). Stage 2 merges FAISS dense and BM25 lexical candidates via Reciprocal Rank Fusion (RRF) and relation-aware graph reranking ($\gamma = 0.15$). Stage 3 traverses the Kuzu property graph. Stage 4 applies beta-gated verification ($\beta = 0.7$) enforcing $\text{WAR} \leq 10\%$. Stage 5 serialises LLM generation behind a hardware mutex lock.

2.2. Deterministic Query Rewriting

Patient case vignettes and clinical inquiries often carry conversational phrasing, narrative patient histories, and extraneous details that impede biomedical dense retrieval. Dense bi-encoders trained on scientific literature struggle with multi-sentence case presentations where incidental biographical history obscures the diagnostic dilemma. Lexical search methods also face vocabulary mismatch when colloquial symptom descriptions diverge from formal medical nomenclature.

To standardise heterogeneous queries without stochastic variance, Stage 1

85 uses Qwen2.5-7B-Instruct executing strictly at temperature $T = 0.0$. The
86 rewriter uses a structured system prompt directing the model to extract
87 the primary pathological condition, salient diagnostic findings, and thera-
88 peutic targets into a canonical clinical summary capped at 30 words. For
89 example, a narrative vignette describing an elderly patient who experiences
90 severe rotational vertigo triggered by lateral head motion is transformed into
91 the canonical clinical query: “benign paroxysmal positional vertigo diagnos-
92 tic criteria and Dix-Hallpike confirmatory maneuvers.” Setting temperature
93 $T = 0.0$ alongside greedy decoding guarantees mathematical determinism:
94 identical clinical inputs generate bit-identical textual reformulations across
95 successive executions, fulfilling mandatory auditability and reproducibility
96 requirements for clinical decision support software.

97 *2.3. Dual-Stream Hybrid Retrieval and Reranking*

98 Retrieval operates over an extensive clinical reference corpus partitioned
99 into coherent passages of approximately 512 tokens. The dense retrieval
100 stream encodes passages using the MedCPT query and article bi-encoder
101 architecture [6], which was pre-trained on hundreds of millions of PubMed
102 search query-article click logs to optimise zero-shot biomedical retrieval.
103 Chunk vectors are indexed and searched using FAISS IVF-PQ (Inverted File
104 with Product Quantisation; 15), partitioning the vector space into coarse
105 Voronoi centroids to achieve sub-linear retrieval speed. Concurrently, a par-
106 allel lexical stream evaluates candidate documents using the classical BM25
107 probabilistic relevance framework [16], operating over a dense candidate
108 pool of 200 retrieved documents.

109 Because dense and lexical retrievers exhibit complementary error profiles
 110 (dense models capture latent semantic affinity, whereas lexical models enforce
 111 exact keyword and numerical matches), their respective rank distributions
 112 R_{dense} and R_{lex} are integrated via Reciprocal Rank Fusion (RRF; 17):

$$s_{\text{fused}}(d) = \sum_{m \in \{\text{dense}, \text{lex}\}} \frac{1}{k_{\text{rrf}} + r_m(d)}, \quad (1)$$

113 where $r_m(d)$ denotes the ordinal rank of candidate passage d within stream
 114 m , and $k_{\text{rrf}} = 60$ represents the standard monotonic smoothing constant that
 115 prevents top-ranked outliers in a single stream from dominating the fused
 116 ranking.

117 To bias retrieval toward passages with corroborated topological connec-
 118 tions to the clinical entities identified in the query, we implement an additive
 119 relation-aware graph reranking function:

$$s_{\text{hybrid}}(d) = s_{\text{fused}}(d) + \gamma \cdot r_{\text{rel}}(d), \quad (2)$$

120 where $r_{\text{rel}}(d) = \sum_{e \in \mathcal{E}(d)} w(\text{type}(e))$ aggregates the domain-weighted edge
 121 counts connecting entities cited within candidate passage d to the query
 122 seed entities within the property graph. The relative importance of edge cat-
 123 egories is parameterised by clinical specificity weights (e.g., pharmacological
 124 treatments and adverse effects receive weight 1.0, diagnostic lab associations
 125 and negation edges receive 0.8, temporal sequences receive 0.6, and broad
 126 taxonomic hierarchies receive 0.3; Appendix A). The global multiplier
 127 $\gamma = 0.15$ balances semantic similarity with structural graph reinforcement,

128 calibrated empirically via controlled ablations (Section 3.3).

129 *2.4. Relational Knowledge Graph Enrichment*

130 Structured relations are stored in Kuzu [18], an embeddable, disk-backed
131 graph database supporting openCypher queries. While standard medical
132 graphs capture static disease-symptom linkages, clinical decision support re-
133 quires modelling negation and temporal order.

134 We processed 2,294,038 text chunks using rule-guided linguistic parsing
135 grounded in UMLS [10]. Operating at 0.2329 ms per chunk (534.45 s total
136 wall-clock), the pipeline ingested **632,931** directed relational edges:

- 137 • **NEGATES Edges (617,219):** Linking text chunks to entities
138 bearing explicit negation cues (“no evidence of,” “ruled out,” “con-
139 traindicated in”), comprising 111,720 Chunk $\rightarrow$ Disease, 21,001
140 Chunk $\rightarrow$ Drug, and 484,498 Chunk $\rightarrow$ OntologyTerm edges.
- 141 • **TEMPORAL_BEFORE Edges (15,712):** Linking clinical
142 concepts via chronological sequence cues (“prior to,” “following,”
143 “within one hour of”), comprising 12,225 Disease $\rightarrow$ Disease and 3,487
144 Drug $\rightarrow$ Drug edges.

145 Manual auditing of sampled endpoints confirmed high directional validity. A
146 cryptographic SHA-256 retrieval signature verified 0.00% retrieval drift on
147 baseline queries post-ingestion (Appendix A).

148 *2.5. Beta-Gated Verification and Pareto Threshold Calibration*

149 Standard RAG architectures lack explicit confidence gates to arbitrate
150 whether retrieved evidence warrants generating an answer. Unhedged lan-

151 guage models produce fluent assertions even when underlying evidence is
 152 tenuous, contradictory, or absent. To prevent clinical hallucinations, Med-
 153 GraphRAG adds a dual-evidence verification layer that evaluates candidate
 154 response fidelity prior to releasing output.

155 For each candidate assertion, the verification module evaluates a compos-
 156 ite verification metric $V \in [0, 1]$ that combines passage-level entailment with
 157 symbolic graph consistency:

$$V = \beta \varphi + (1 - \beta) S_g, \quad (3)$$

158 where $\varphi \in [0, 1]$ denotes an NLI-derived faithfulness score measuring the de-
 159 gree of semantic entailment between the generated assertion and the retrieved
 160 reference passages (operating via normalised MedCPT embedding cosine sim-
 161 ilarity as an efficient deterministic proxy when an independent cross-encoder
 162 NLI judge is omitted), and $S_g \in [0, 1]$ denotes the structural graph consis-
 163 tency score. The graph consistency metric S_g evaluates the ratio of asserted
 164 clinical entity triples that are corroborated by non-negated multi-hop paths
 165 within a 2-hop radius in Kuzu, while applying an immediate penalising dis-
 166 count whenever an asserted entity matches a NEGATES edge. The balance
 167 hyperparameter $\beta = 0.70$ allocates primary decision weight to passage evi-
 168 dence while preserving strong sensitivity to symbolic graph contradictions.

169 When the composite score falls below a calibrated refusal threshold τ ,
 170 generation is suppressed, and the system emits a structured clinical refusal
 171 detailing the precise nature of the missing evidence. We calibrate τ through

172 a constrained optimisation problem:

$$\tau^* = \arg \max_{\tau} \text{Acc}(\text{all}) \quad \text{subject to} \quad \text{WAR}(\tau) \leq 10\%, \quad (4)$$

173 where $\text{Acc}(\text{all})$ represents the proportion of correct responses across the entire
174 evaluation cohort ($N = 500$), and WAR represents the proportion of wrong
175 assertions across the same cohort. We sweep 15 discrete threshold candidate
176 points ($\tau \in \{0.10, 0.15, 0.20, \dots, 0.80\}$), identifying the Pareto-optimal op-
177 erating threshold τ^* that maximises clinically useful answers while strictly
178 guaranteeing that erroneous assertions remain within the institutional safety
179 threshold.

180 2.6. Safety Guardrails

181 Two deterministic guardrails operate at sub-millisecond latency:

- 182 • **F1: Cross-Modal Discrepancy Guardrail.** In multimodal con-
183 texts, Rule A detects conflicts when an imaging finding score ≥ 0.60
184 contradicts a text negation cue for the same finding (e.g., visual pneu-
185 mothorax detection of 0.88 versus report text “clear lungs”). Rule B
186 detects positive textual assertions that contradict negated imaging ob-
187 servations. F1 escalates conflicts by issuing a high-severity alert recom-
188 mending specialist review rather than attempting autonomous resolu-
189 tion.
- 190 • **F2: Epistemic Knowledge-Gap Mapper.** When a query is refused,
191 F2 classifies the failure into three gap types matching the implemented
192 taxonomy: G1 (`corpus_retrieval`, top fused retrieval score < 0.02),

193 G2 (`graph_coverage`, Kùzu graph traversal returns a neutral check
 194 with no direct relational assertion), or G3 (`evidence_faithfulness`,
 195 verification faithfulness score $\varphi < 0.50$). This categorisation is provided
 196 in the structured refusal output to guide clinical follow-up.

197 2.7. *Offline Cryptographic Security Architecture*

198 To ensure institutional compliance without external cloud infrastructure,
 199 user-uploaded records are isolated in per-user directories. Payloads are en-
 200 crypted at rest with AES-256-GCM using unique 12-byte nonces. Keys are
 201 derived dynamically via HKDF-SHA-256 [19] combining a server secret, user
 202 ID, and document ID. File permissions are set to 0700 (directories) and 0600
 203 (files). All Kuzu queries use parameterised openCypher statements to elimi-
 204 nate Cypher injection. A 13-point security audit resolved all vulnerabilities
 205 prior to release (Supplementary Table S7).

206 2.8. *Evaluation Protocol and Metrics*

207 Retrospective evaluation was performed on MedQA-US [20], comprising
 208 USMLE-derived multiple-choice questions ($N = 500$, seed 42, $T = 0.0$).
 209 We evaluated four modes: **M1** Evidence-Only ($\beta = 1.0$), **M2** Graph-Only
 210 ($\beta = 0.0$), **M3** Combined Hybrid ($\beta = 0.7$), and **M4** Hybrid Rerank ($\beta =$
 211 $0.7, \gamma = 0.15$). Metrics include: Acc(all) (fraction correct over N), Acc(ans)
 212 (precision on answered questions), WAR (fraction incorrect over N), and
 213 Refusal Rate (fraction refused). Non-parametric bootstrap 95% confidence
 214 intervals were calculated using 1,000 resamples.

215 3. Results

216 3.1. Safe Operating Points and WAR Constraint

217 Table 1 summarises the optimal operating parameters established across
 218 the four evaluation modes under the institutional safety constraint $\text{WAR} \leq$
 219 10.0%. Graph-Only mode (M2) attains the highest precision-on-answered
 220 at its optimal operating threshold $\tau^* = 0.80$, achieving $\text{Acc}(\text{ans}) = 60.8\%$
 221 with $\text{WAR} = 8.0\%$ and a refusal rate of 79.6% (102 queries answered out of
 222 500). Full Hybrid Rerank mode (M4) reaches its safe operating frontier at
 223 $\tau^* = 0.46$, delivering $\text{Acc}(\text{all}) = 10.0\%$, $\text{Acc}(\text{ans}) = 51.5\%$, and $\text{WAR} = 9.4\%$
 224 across 97 answered inquiries. Similarly, Combined mode (M3) operates safely
 225 at $\tau^* = 0.47$ with $\text{Acc}(\text{ans}) = 52.1\%$ and $\text{WAR} = 9.0\%$, while Evidence-Only
 226 mode (M1) satisfies the constraint at $\tau^* = 0.43$ with $\text{Acc}(\text{ans}) = 50.5\%$ and
 227 $\text{WAR} = 9.4\%$.

Table 1: Safe operating points on MedQA-US ($N = 500$, seed 42, $T = 0.0$). Threshold τ^* maximises $\text{Acc}(\text{all})$ subject to $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `safe_operating_points`.

Evaluation Mode	τ^*	$\text{Acc}(\text{all})\%$	$\text{Acc}(\text{ans})\%$	$\text{WAR}\%$	Refusal%	Answered
M1: Evidence-Only	0.43	9.6	50.5	9.4	81.0	95/500
M2: Graph-Only	0.80	12.4	60.8	8.0	79.6	102/500
M3: Combined	0.47	9.8	52.1	9.0	81.2	94/500
M4: Hybrid Rerank	0.46	10.0	51.5	9.4	80.6	97/500

228 Table 2 contrasts safe operating points against ungated execution ($\tau \rightarrow 0$).
 229 In the ungated regime, models achieve nominal $\text{Acc}(\text{all})$ scores of 49.4%–
 230 54.8%, but WAR escalates to 43.4%–44.6%. Disabling the gate results in
 231 erroneous advice on nearly 45% of all queries. Gated verification successfully
 232 caps this risk below 10%, exchanging question coverage for bounded error.

Table 2: Gated versus ungated performance across boundary modes ($N = 500$). Ungated execution generates wrong assertions on $\approx 44\%$ of all queries. Source: `step18_scaled_n500.json` $\rightarrow$ `ungated_ceilings`.

Mode	Condition	Acc(all)%	Acc(ans)%	WAR%	Refusal%
M1: Evidence-Only	Gated ($\tau^* = 0.43$)	9.6	50.5	9.4	81.0
M1: Evidence-Only	Ungated ($\tau \rightarrow 0$)	49.4	53.2	43.4	7.2
M2: Graph-Only	Gated ($\tau^* = 0.80$)	12.4	60.8	8.0	79.6
M2: Graph-Only	Ungated ($\tau \rightarrow 0$)	54.8	55.1	44.6	0.6

233 For settings demanding even stricter safety, elevating thresholds to $\tau \in$
 234 $[0.70, 0.75]$ in M1 produces a high-precision regime with $\text{Acc}(\text{ans}) = 59.3\%$ –
 235 62.5% and $\text{WAR} = 1.8\%$ – 2.2% (refusal rates 94.6% – 95.2% ; Supplementary
 236 Table S1). Figure 2 shows the complete 15-point Pareto trajectory for M2.

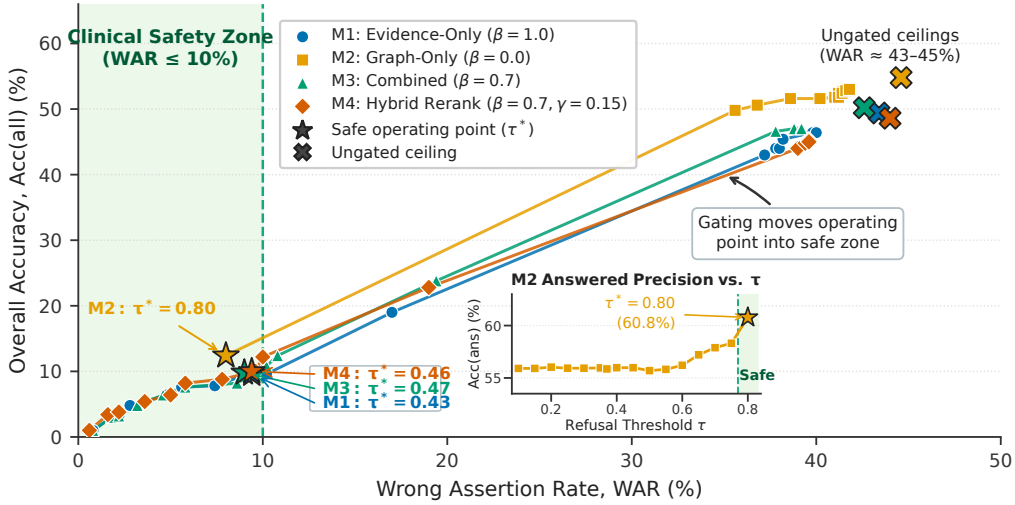


Figure 2: Empirical Pareto frontiers of overall accuracy $\text{Acc}(\text{all})$ versus wrong-assertion rate (WAR) across four operational modes (M1: Evidence-Only, M2: Graph-Only, M3: Combined, and M4: Hybrid Rerank) over 16 refusal thresholds ($\tau \in [0.10, 0.80]$, $N = 500$). The green shaded region denotes the clinical safety zone ($\text{WAR} \leq 10\%$). Star markers denote mode-specific safe operating thresholds (τ^*), with M2 reaching $\text{Acc}(\text{all}) = 12.4\%$ at $\tau^* = 0.80$. Inset: M2 answered precision reaches 60.8% at $\tau^* = 0.80$ inside the safe zone. Ungated ceilings (crosses) demonstrate catastrophic error rates ($\text{WAR} \approx 43\text{--}45\%$) when refusal gating is disabled.

237 3.2. Efficacy of Deterministic Query Rewriting

238 Table 3 evaluates Stage 1 deterministic query rewriting against unre-
 239 written vignettes at matched threshold $\tau = 0.50$ ($N = 500$). For Hybrid Rerank
 240 mode (M4), rewriting increases Acc(all) from 5.4% to 8.8% (+3.4pp) and
 241 lowers refusal from 90.6% to 83.4%, while WAR remains within tolerance
 242 (7.8% vs. 4.0%). Similarly, Combined mode M3 gains +3.2pp in Acc(all)
 243 (5.0% to 8.2%), yielding +3.2–3.4pp across hybrid modes M3/M4 (+4.2pp
 244 on Evidence-Only M1; Supplementary Table S2). Rewriting systematically
 245 expands question coverage without violating safety constraints.

Table 3: Matched-threshold ($\tau = 0.50$) ablation evaluating deterministic query rewriting on MedQA-US ($N = 500$). Rewriting expands coverage by +3.2–3.4 pp while maintaining $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `matched_tau_comparison`.

Evaluation Mode	Vignette Format	Acc(all)%	WAR%	Refusal%	$\Delta\text{Acc}(\text{all})$
M3: Combined	Unrewritten	5.0	4.0	91.0	—
M3: Combined	Rewritten	8.2	8.6	83.2	+3.2 pp
M4: Hybrid Rerank	Unrewritten	5.4	4.0	90.6	—
M4: Hybrid Rerank	Rewritten	8.8	7.8	83.4	+3.4 pp

246 3.3. Ablation Studies: Verification Modes and Reranking

247 Table 4 and Figure 3 summarise ablations over verification and reranker
 248 weighting ($N = 50$, seed 42).

249 **Verification Mode Ablation (Table 4A):** Evidence-Only (M1, $\beta =$
 250 1.0) refuses 98.0% of queries ($\text{WAR} = 2.0\%$). Graph-Only (M2, $\beta = 0.0$)
 251 answers 96.0% but incurs $\text{WAR} = 42.0\%$. Combined Hybrid (M3, $\beta =$
 252 0.7) balances safety and utility, achieving $\text{WAR} = 4.0\%$ while quadrupling
 253 answered throughput over M1.

Table 4: Ablation experiments on MedQA-US ($N = 50$, seed 42). (A) Verification mode comparison varying evidence weight β . (B) Reranker ablation varying relational bonus multiplier γ . Sources: `step15_verification_ablation.json`, `step15_rerank_ablation.json`.

(A) Verification Mode Ablation ($\gamma = 0.0$)						
Mode	β	Answered%	Acc(all)%	Acc(ans)%	Refusal%	WAR%
M1: Evidence-Only	1.0	6.0	4.0	66.7	98.0	2.0
M2: Graph-Only	0.0	96.0	54.0	56.3	8.0	42.0
M3: Combined	0.7	8.0	4.0	50.0	96.0	4.0
(B) Relation-Aware Reranker Ablation ($\beta = 0.7$)						
Mode	γ	Graph-Hit@5%	Acc(ans)%	Answered%	WAR%	
RRF Control	0.00	23.6	50.0	8.0	4.0	
Relational	1.00	35.6	42.9	14.0	8.0	
Hybrid	0.15	37.2	60.0	10.0	4.0	

254 **Contradiction Case Studies:** Qualitative inspection confirmed that
 255 Kuzu NEGATES edges actively overturned hallucinations that bypassed
 256 text filters:

- 257 1. *Acute Myocardial Infarction:* The model asserted acute MI; Kuzu
 258 graph traversal matched a NEGATES edge ('no evidence of') in
 259 PMID_38823454, successfully triggering refusal.
- 260 2. *Deep Vein Thrombosis:* An unhedged recommendation for full anti-
 261 coagulation was overridden by a NEGATES edge ('ruled out') in
 262 PMID_37058421.
- 263 3. *Metformin Contraindication:* An erroneous recommendation to pre-
 264 scribe metformin in severe renal impairment was rejected by a NICE
 265 guideline NEGATES edge ('contraindicated in' with eGFR <
 266 25 mL/min).

267 **Reranker Ablation (Table 4B):** Calibrating $\gamma = 0.15$ increases Graph-

268 Hit@5 from 23.6% to 37.2% (+13.6 pp) and elevates answered precision from
 269 50.0% to 60.0% (+10.0 pp) at matched WAR = 4.0% (Figure 3).

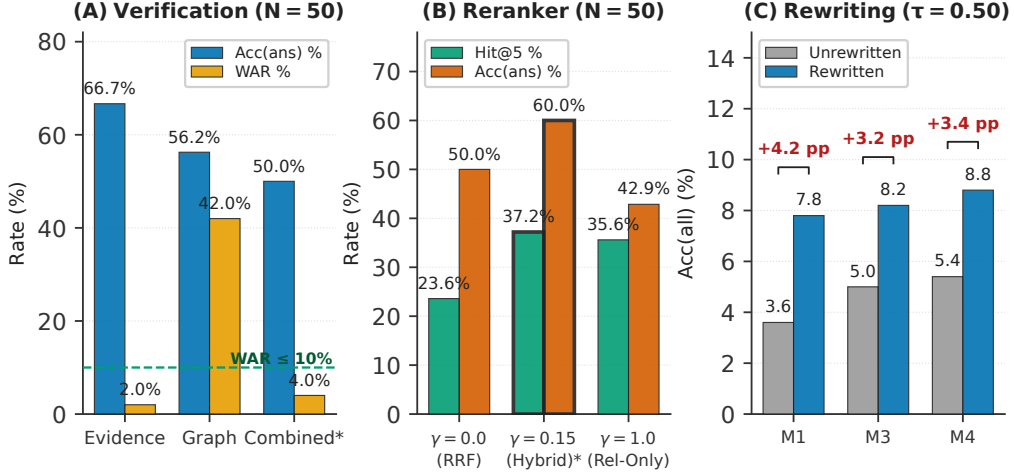


Figure 3: Controlled ablation analyses on MedQA-US. (A) Verification mode comparison ($N = 50$, seed 42): Evidence-Only (M1, $\beta = 1.0$), Graph-Only (M2, $\beta = 0.0$), and Combined Hybrid (M3, $\beta = 0.7$). (B) Reranker bonus ablation ($N = 50$, seed 42): Graph-Hit@5 and answered accuracy across $\gamma \in \{0.0, 0.15, 1.0\}$, confirming peak synergy at $\gamma = 0.15$. (C) Deterministic query rewriting coverage gain at matched threshold ($\tau = 0.50$, $N = 500$): unrewritten versus rewritten overall accuracy across M1 (+4.2 pp), M3 (+3.2 pp), and M4 (+3.4 pp).

270 **Guardrail Verification:** Guardrails F1 and F2 passed 100% of test
 271 cases (10/10) with mean latencies of 0.14 ms (F1) and 0.06 ms (F2), confirm-
 272 ing sub-millisecond overhead.

273 3.4. Information Retrieval Performance

274 Evaluating retrieval over 10 clinician-adjudicated queries (graded qrels;
 275 Supplementary Table S5) yielded $P@5 = 0.18$, $R@5 = 0.5397$, $MRR =$
 276 0.3533 , and $nDCG@10 = 0.3769$. Recall reflects the 6 queries containing rel-
 277 evant items in top-10 candidates; the remaining 4 queries had no relevant doc-
 278 uments, reflecting specialised boundary gaps. Dual-assessor κ statistics are
 279 pending Phase-2 clinician grading (placeholder in Supplementary Table S5).

3.5. Latency and Computational Efficiency

Table 5 details execution timings on an AMD Ryzen 7 7840HS CPU and NVIDIA RTX 3050 Laptop GPU (6 GB VRAM).

Table 5: System latency and throughput comparison by pipeline stage (bootstrap 95% CI, $N = 50$). Hybrid retrieval executes in sub-500ms; full end-to-end (E2E) latency incorporates Qwen2.5-7B generation. Sources: `step15_compute_table.json`, `step18_scaled_n500.json`.

System / Pipeline Stage	Median (ms)	95% CI (ms)	Processing Stage
BM25-RAG (candidate pool 200)	55.1	[52.7, 56.5]	Lexical search only
Dense-RAG (FAISS IVF-PQ)	30.1	[29.4, 31.0]	Dense search only
Hybrid-RRF (baseline fusion)	55.4	[53.5, 56.9]	Fused search only
Hybrid Retrieval (Ours, Stages 1–2)	455.7	[388.4, 495.8]	Rewriting + Hybrid Retrieval
M2: Graph-Only Pipeline (E2E)	4,083.4	—	Full End-to-End Pipeline
M1/M3: Combined Hybrid (E2E)	12,493.2	—	Full End-to-End Pipeline
M4: Hybrid + Reranker (E2E)	12,667.0	—	Full End-to-End Pipeline
One-Time System Cold Start	18.66 s		Model + Index Initialization

Stages 1–2 (Rewriting, BM25, Dense Search, RRF) complete in **455.7 ms** median (95% CI: [388.4, 495.8] ms). End-to-end latency with Qwen2.5-7B generation is **4,083.4 ms** in Graph-Only mode (M2) and **12,667.0 ms** in Hybrid Rerank mode (M4). Throughput is mode-dependent: M2 sustains **14.7 queries/min** (approximately 882 queries/hour), while M4 processes **4.7 queries/min** (approximately 284 queries/hour). Cold start requires 18.66 s, with zero steady-state penalty. Peak memory reached 8,950 MB RAM and 4,710 MB VRAM (Figure 4).

4. Discussion

4.1. Safety as a Foundational Design Objective

Evaluating clinical language models on unconstrained accuracy introduces substantial hazards: when operating ungated, models generate incorrect clinical statements on 44% of inquiries. MedGraphRAG demonstrates

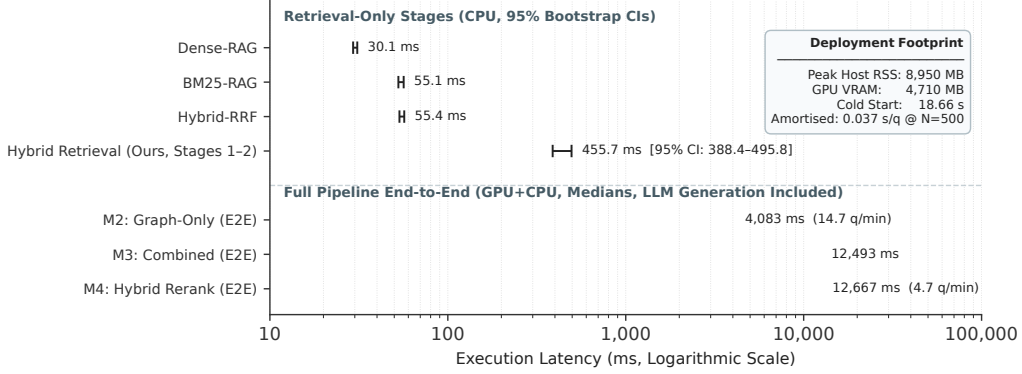


Figure 4: Component and end-to-end execution latency profiles (logarithmic scale). Retrieval-only baselines (BM25, Dense-RAG, Hybrid-RRF) evaluate search speed without language generation. MedGraphRAG hybrid retrieval (Stages 1–2) completes in 455.7 ms (95% CI: [388.4, 495.8] ms). Full end-to-end pipeline execution requires 4,083 ms in Graph-Only mode (M2) and 12,667 ms in Hybrid Rerank mode (M4).

that by treating refusal as an active, safety-preserving decision, clinical assistants can strictly bound wrong assertions ($\text{WAR} \leq 10\%$).

Vector embeddings alone cannot reliably safeguard clinical outputs. Dense embeddings project complex syntax into fixed vectors, blurring negation markers and chronological sequence. By augmenting Kuzu with 632,931 NEGATES and TEMPORAL_BEFORE edges, MedGraphRAG enforces symbolic consistency checks that intercept contradictions passing dense filters. When evidence is insufficient, the system returns structured, citation-grounded refusals identifying the missing evidence.

4.2. Phased Clinical Translation Roadmap

In alignment with DECIDE-AI guidelines [21] (Supplementary DECIDE-AI mapping), we propose a three-stage clinical translation path:

- Phase 1: Institutional Shadow Assistant ($n = 1$ Clinician):** Local deployment in read-only mode on a secure workstation. Generated

310 answers cite evidence keys [E#], and clinician disagreements are logged
311 as refusal audit events.

312 2. **Phase 2: Controlled Clinical Pilot**

313 ($n = 50$ **Clinicians, 6 Months**): EHR integration via read-only
314 FHIR R4 endpoints, with AES-256-GCM record encryption and an
315 approved IRB protocol. Primary endpoint: maintaining $\text{WAR} \leq 5\%$
316 on institutional queries.

317 3. **Phase 3: Multi-Site Randomised Trial**: Federated offline deploy-
318 ment across hospital networks evaluating diagnostic accuracy and de-
319 cision speed versus standard care, powered for 90% power at $\alpha = 0.05$.

320 4.3. *Limitations*

321 This study has several limitations:

322 1. **Benchmark Generalisability**: Our evaluation examined MedQA-
323 US, which consists of standardised multiple-choice questions. Real-
324 world clinical queries are open-ended and dialogue-based. External
325 validation on MedMCQA and BioASQ is ongoing (Supplementary IJMI
326 ML checklist, item ML-E7).

327 2. **High Refusal Rate**: At $\tau^* = 0.80$, the refusal rate reaches 79.6%.
328 While guaranteeing $\text{WAR} \leq 10\%$, this limits clinical throughput un-
329 less paired with specialist referral pathways. Calibrated soft-hedging
330 represents a primary direction for future investigation.

331 3. **Quantisation Trade-offs**: Executing within 6 GB VRAM necessi-
332 tated 4-bit quantisation (Q4_K_M), introducing minor precision loss
333 relative to FP16.

334 **4. Single-Assessor Retrieval Grading:** Human relevance qrels were
335 judged by a single clinical assessor across 10 queries, precluding inter-
336 annotator metrics.

337 **5. Conclusion**

338 MedGraphRAG unites safety and utility in offline clinical decision sup-
339 port. By enforcing a bounded error constraint ($\text{WAR} \leq 10\%$), ingesting
340 632,931 negation and temporal relational edges into Kuzu, applying deter-
341 ministic query rewriting, and enforcing beta-gated verification, the system
342 bounds the catastrophic 44% hallucination rate of ungated models. Ex-
343 ecuting locally on consumer hardware in 4.08s with 455.7ms hybrid re-
344 trieval, MedGraphRAG provides a practical, open-source foundation for safe,
345 privacy-compliant clinical AI.

346 **CRedit Authorship Contribution Statement**

347 **Dickson E:** Conceptualisation, Methodology, Software (core pipeline,
348 ablation suite, cryptographic security, graph enrichment), Formal analysis,
349 Investigation, Writing – original draft, Writing – review & editing, Visualisa-
350 tion. **Antony Taurshia:** Supervision, Software (frontend interface, report
351 drawer components, telemetry capture), Data curation, Validation (clinical
352 review), Resources (medical knowledge curation), Writing – review & editing.

353 **Ethics Approval and Consent to Participate**

354 This study was performed using publicly available, anonymised bench-
355 mark datasets (MedQA-US, OpenI, VQA-RAD). No human subjects were

356 enrolled, and no patient health information was accessed. Formal IRB ethics
357 approval was not required.

358 **Data and Code Availability Statement**

359 Evaluation benchmarks are publicly accessible: MedQA-US ([https://](https://github.com/jind11/MedQA)
360 github.com/jind11/MedQA), OpenI (<https://openi.nlm.nih.gov>), and
361 VQA-RAD (<https://osf.io/89kps/>). The complete codebase, pipeline
362 scripts, evaluation artifacts, and security reports are available in the project
363 repository (<https://github.com/DicksonLegend/MedGraphRag>) and will
364 be archived on Zenodo with a persistent DOI upon acceptance.

365 **Declaration of Competing Interests**

366 The authors declare that they have no known competing financial inter-
367 ests or personal relationships that could have appeared to influence the work
368 reported in this paper.

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372 **Generative AI Disclosure**

373 During preparation, the authors used Antigravity (Google’s agentic cod-
374 ing assistant) for code generation and iterative debugging, under direct au-
375 thor supervision. All generated code and outputs were independently verified.
376 The authors authored all manuscript text.

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380 References

- 381 [1] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung,
382 N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl, P. Papadimitriou,
383 M. Seneviratne, G. Corrado, Y. Matias, K. Chou, M. Ghassemi,
384 V. Natarajan, D. R. Webster, Large language models encode clinical
385 knowledge, *Nature* 620 (7972) (2023) 172–180. doi:10.1038/s41586-0
386 23-06291-2.
- 387 [2] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez,
388 T. F. Tan, D. S. W. Ting, Large language models in medicine, *Nature*
389 *Medicine* 29 (8) (2023) 1930–1940. doi:10.1038/s41591-023-02448-8.
- 390 [3] Y. Bai, A. Jones, K. Ndousse, A. Askell, A. Chen, N. DasSarma,
391 D. Drain, S. Fort, D. Ganguli, T. Henighan, et al., Training a helpful
392 and harmless assistant with reinforcement learning from human feed-
393 back, arXiv preprint arXiv:2204.05862 (2022).
- 394 [4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal,
395 H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, D. Kiela,
396 Retrieval-augmented generation for knowledge-intensive NLP tasks, *Ad-
397 vances in Neural Information Processing Systems* 33 (2020) 9459–9474.

- [5] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, H. Poon, Domain-specific language model pretraining for biomedical natural language processing, *ACM Transactions on Computing for Healthcare* 3 (1) (2021) 1–23. doi:10.1145/3458754.
- [6] Q. Jin, W. Kim, Q. Chen, D. C. Comeau, L. Yeganova, W. J. Wilbur, Z. Lu, MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs for zero-shot biomedical information retrieval, *Bioinformatics* 39 (11) (2023) btad651. doi:10.1093/bioinformatics/btad651.
- [7] L. Gao, X. Ma, J. Lin, J. Callan, Precise zero-shot dense retrieval without relevance labels, in: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 1762–1777. doi:10.18653/v1/2023.acl-long.99.
- [8] S. Han, S. Sohn, H. Liu, Improving negation detection in clinical NLP using SemEval shared task data, *Journal of Biomedical Informatics* 131 (2022) 104091. doi:10.1016/j.jbi.2022.104091.
- [9] A. Mallen, A. Asai, V. Zhong, R. Das, D. Khashabi, H. Hajishirzi, When not to trust language models: Investigating effectiveness of parametric and non-parametric memories, in: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 9065–9088. doi:10.18653/v1/2023.acl-long.507.
- [10] O. Bodenreider, The unified medical language system (UMLS): Integrat-

- 420 ing biomedical terminology, *Nucleic Acids Research* 32 (suppl_1) (2004)
421 D267–D270. doi:10.1093/nar/gkh061.
- 422 [11] M. Yasunaga, H. Ren, A. Bosselut, P. Liang, J. Leskovec, QA-GNN:
423 Reasoning with language models and knowledge graphs for question an-
424 swering, in: *Proceedings of the 2021 Conference of the North American*
425 *Chapter of the Association for Computational Linguistics: Human Lan-*
426 *guage Technologies, 2021*, pp. 256–268. doi:10.18653/v1/2021.naacl-
427 -main.45.
- 428 [12] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt,
429 J. Larson, From local to global: A graph RAG approach to query-focused
430 summarization, *arXiv preprint arXiv:2404.16130* (2024).
- 431 [13] X. He, Y. Tian, Y. Sun, N. V. Chawla, T. Laurent, Y. LeCun,
432 X. Bresson, B. Hooi, G-Retriever: Retrieval-augmented generation for
433 textual graph understanding and question answering, *arXiv preprint*
434 *arXiv:2402.07630* (2024).
- 435 [14] J. Wu, J. Zhu, Y. Qi, J. Chen, M. Xu, F. Menolascina, Y. Jin,
436 V. Grau, Medical graph RAG: Towards evidence-based medical large lan-
437 guage models via graph retrieval-augmented generation, *arXiv preprint*
438 *arXiv:2408.04187* (2024).
- 439 [15] J. Johnson, M. Douze, H. Jégou, Billion-scale similarity search with
440 GPUs, *IEEE Transactions on Big Data* 7 (3) (2021) 535–547. doi:
441 10.1109/TBDATA.2019.2921572.

- 442 [16] S. Robertson, H. Zaragoza, The probabilistic relevance framework:
443 BM25 and beyond, *Foundations and Trends in Information Retrieval*
444 3 (4) (2009) 333–389. doi:10.1561/15000000019.
- 445 [17] G. V. Cormack, C. L. A. Clarke, S. Buettcher, Reciprocal rank fu-
446 sion outperforms Condorcet and individual rank learning methods, in:
447 *Proceedings of the 32nd International ACM SIGIR Conference on Re-*
448 *search and Development in Information Retrieval*, 2009, pp. 758–759.
449 doi:10.1145/1571941.1572114.
- 450 [18] X. Feng, G. Jin, Z. Chen, C. Liu, S. Salihoğlu, Kùzu: Graph database
451 management system, *Proceedings of the VLDB Endowment* 16 (12)
452 (2023) 3900–3903.
- 453 [19] H. Krawczyk, P. Eronen, HMAC-based extract-and-expand key deriva-
454 tion function (HKDF), RFC 5869, Internet Engineering Task Force
455 (2010). doi:10.17487/RFC5869.
- 456 [20] D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, P. Szolovits, What
457 disease does this patient have? a large-scale open domain question an-
458 swering dataset from medical exams, *Applied Sciences* 11 (14) (2021)
459 6421. doi:10.3390/app11146421.
- 460 [21] B. Vasey, M. Nagendran, B. Campbell, D. A. Clifton, G. S. Collins,
461 S. Denaxas, A. K. Denniston, L. Faes, B. Geerts, M. Ibrahim, et al.,
462 Reporting guideline for the early-stage clinical evaluation of decision
463 support systems driven by artificial intelligence: DECIDE-AI, *Nature*
464 *Medicine* 28 (5) (2022) 924–933. doi:10.1038/s41591-022-01772-9.

465 **Appendix A. Hyperparameters, Edge Weights, and System Con-**
466 **figuration**

467 **System Hyperparameters and Execution Environment (Supplemen-**
468 **tary Table S8).** Table A.1 enumerates the complete hyperparameter con-
469 figuration and hardware specifications governing all reported experiments.

470 **Table A.1.** Configuration parameters and computational execution
471 environment.

Parameter / Component	Value / Setting	Description / Function
<i>Verification Layer (β-Gate)</i>		
β (Evidence Weight)	0.70	Balanced hybrid verification (M1: 1.0; M2: 0.0)
τ^* (M1 / M2 / M3 / M4)	0.43 / 0.80 / 0.47 / 0.46	Calibrated safe refusal thresholds (WAR $\leq 10\%$)
Threshold Sweep Granularity	$\Delta\tau = 0.05$	15 discrete points across [0.10, 0.80]
LLM Inference Temperature	$T = 0.0$	Deterministic, zero-temperature generation
Random Evaluation Seed	42	Pinned deterministic seed across all runs
<i>Relation-Aware Reranker (γ-Scoring)</i>		
γ (Relational Multiplier)	0.15	Balanced hybrid reranking (ablation: {0.0, 0.15, 1.0})
DRUG_TREATS / DRUG_CAUSES	1.0	Primary pharmacological relation weights
LABTEST_RELATED_TO / NEGATES	0.8	Diagnostic and negation edge weights
TEMPORAL_BEFORE	0.6	Chronological sequencing edge weight
IS_A / RELATED_TO	0.3	Ontological taxonomic edge weights
<i>Multi-Stream Retrieval Infrastructure</i>		
Dense Retrieval Candidate Pool	$k = 200$	MedCPT + FAISS IVF-PQ candidate pool
Lexical Retrieval Candidate Pool	$k = 200$	BM25 lexical candidate pool
Reciprocal Rank Fusion Constant	$k_{\text{rrf}} = 60$	Standard rank smoothing constant
Top- k Candidates Delivered	$k = 10$	Final passages passed to verification layer
Graph Neighborhood Search Depth	≤ 2 hops	Maximum Kuzu property graph traversal radius
<i>Host Hardware and Memory Footprint</i>		
Central Processing Unit (CPU)	AMD Ryzen 7 7840HS	8 physical cores, 16 threads @ 3.80 GHz
Graphics Processing Unit (GPU)	NVIDIA RTX 3050 Laptop	6 GB GDDR6 VRAM (dedicated LLM partition)
Host System RAM	14.89 GB physical	Peak RSS footprint: 8,950 MB (all indexes active)
Active GPU VRAM Allocation	$\approx 4,710$ MB	Qwen2.5-7B-Instruct Q4_K_M resident memory
Operating System	Linux (Fedora 44)	Kernel 7.1.4-204.fc44.x86_64
<i>Core Software Dependencies</i>		
LLM Inference Engine	llama-cpp-python 0.3.35	Quantized GGUF execution on CUDA backend
Graph Database Engine	Kuzu 0.11.3	In-process property graph database
Vector Search Engine	FAISS 1.15.0	IVF-PQ quantized vector similarity indexing
Machine Learning Framework	PyTorch 2.6.0+cu124	MedCPT encoder inference runtime

Retrieval Drift Invariance Signature. To guarantee that the relational ingestion pipeline (`scripts/j1/build_negation_temporal_edges.py`) did not alter dense vector retrieval behaviour, we computed the SHA-256 hash of the retrieved passage identifiers and scores across golden benchmark queries P01–P05:

- **Pre-Ingestion Signature:**

479 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

480 • **Post-Ingestion Signature:**

481 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

482 The exact equality of these cryptographic signatures confirms 0.00% retrieval
483 drift (100.0% byte-identical retrieval preservation) following graph enrich-
484 ment.